Number Relationships

Number · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

A+ Maths Hub

Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 66 minutes · Total marks: 66

KEY RULES

Fractions of an amount	Divide by the denominator, multiply by the numerator
Percentages	Multiplier method: increase 12% → × 1.12 ; decrease 12% → × 0.88
Ratio	Share in a ratio: total parts, one part, then multiply
Fractions	Add / subtract with a common denominator; multiply tops and bottoms
Reverse percentages	Divide by the multiplier

WORKED EXAMPLE 1

[2 marks]

Work out $\frac{3}{5}$ **of** 85.

Step 1	Divide by the denominator	$85 \div 5 = 17$
Step 2	Multiply by the numerator	$17 \times 3 = 51$

Answer: 51

WORKED EXAMPLE 2

[3 marks]

Increase 240 **by** 15%.

Step 1	The multiplier for a 15% increase	$1 + 0.15 = 1.15$
Step 2	Multiply	$240 \times 1.15 = 276$

Answer: 276

WORKED EXAMPLE 3

[3 marks]

Share 84 **in the ratio** 2 : 5.

Step 1	Add the parts	$2 + 5 = 7$
Step 2	Value of 1 part	$84 \div 7 = 12$
Step 3	Multiply each part	$2 \times 12 = 24$, $5 \times 12 = 60$

Answer: 24 **and** 60

A+ Maths Hub International Community Schools

1 Show that $\frac{3}{7}+\frac{1}{4}=\frac{19}{28}$

(Total for question = 2 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2025 · Question 13

2 Use algebra to show that $0.38\dot{1} = \frac{21}{55}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2023 · Question 13

3 Use algebra to show that $0.7\dot{6}\dot{3} = \frac{42}{55}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2024 · Question 15

4 Use algebra to show that $0.3\dot{7}\dot{8} = \frac{25}{66}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2025 · Question 16

5 Use algebra to show that the recurring decimal $0.6\dot{1}\dot{2} = \frac{101}{165}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2025 · Question 16

A* Maths Hub International Community Schools

6

Use algebra to show that $0.1\dot{7}\dot{6} = \frac{35}{198}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2023 · Question 16

7

Use algebra to show that $0.4\dot{3}\dot{8} = \frac{217}{495}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 16

8

Use algebra to show that $0.5\dot{3}\dot{9} = \frac{89}{165}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2026 · Question 17

9

Use algebra to show that $0.3\dot{4}\dot{5} = \frac{19}{55}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2022 · Question 17

10

Use algebra to show that $0.\dot{7}0\dot{2} = \frac{26}{37}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 18

A+ Maths Hub International Community Schools

11 Use algebra to show that $0.\dot{3}0\dot{6} = \frac{34}{111}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2024 · Question 18

12 (a) Write 75 as a fraction of 125

Give your fraction in its simplest form.

.....

(1)

(b) Write $\frac{21}{105}$ as a percentage.

.....%

(1)

(c) Write 4% as a decimal.

.....

(1)

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 8

13 Show that $4\frac{2}{3} \div 1\frac{5}{6} = 2\frac{6}{11}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2023 · Question 1

A+ Maths Hub International Community Schools

14 Show that $3\frac{5}{7} \div 1\frac{5}{8} = 2\frac{2}{7}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 1

15 Show that $3\frac{1}{5} \times 1\frac{5}{6} = 5\frac{13}{15}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2021 · Question 1

A+ Maths Hub
International Community Schools

16

Show that $6\frac{3}{4} \div 2\frac{4}{7} = 2\frac{5}{8}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2022 · Question 2

17

Show that $2\frac{4}{7} \div 1\frac{1}{8} = 2\frac{2}{7}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2021 · Question 2

18

Show that $2\frac{4}{7} \times 3\frac{1}{9} = 8$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2024 · Question 4

A+ Maths Hub International Community Schools

19 Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2022 · Question 4

CHALLENGE QUESTIONS

20 Show that $2\frac{4}{5} \times 3\frac{3}{7} = 9\frac{3}{5}$

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2026 · Question 17

21 $0.4\dot{x}$ is a recurring decimal.
x is a whole number such that $1 \leqslant x \leqslant 9$

Find, in terms of x, the recurring decimal $0.4\dot{x}$ as a fraction.
Give your fraction in its simplest form.
Show clear algebraic working.

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2021 · Question 18

A* Maths Hub
International Community Schools

22 3 A box contains only

9 red bricks
43 blue bricks
and some yellow bricks

$\frac{7}{20}$ of the bricks are yellow bricks.

Each brick weighs 35 grams.

Work out the total weight of the yellow bricks.

..... grams

(Total for Question 3 is 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2025 · Question 3

23 (a) Use algebra to show that $0.3\dot{7}\dot{2} = \frac{41}{110}$

(2)

b) Express $\frac{\sqrt{125} + \sqrt{80}}{\sqrt{3}}$ in the form $\sqrt{n}$ where n is an integer.

Show your working clearly.

.....

(3)

(Total for Question 15 is 5 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 15

A+ Maths Hub International Community Schools

24 (a) Use algebra to show that $4.\dot{5}\dot{7} = 4\frac{19}{33}$

(2)

(b) Show that $\frac{2}{6-3\sqrt{2}}$ can be written in the form $\frac{a+\sqrt{a}}{b}$

where a and b are integers.

Show your working clearly.

(3)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2021 · Question 15

TOTAL FOR PAPER IS 66 MARKS

A+ Maths Hub International Community Schools